Homework 1

🧮 Question 1: Evaluate the following expressions

(Use integer division for all / operations.)

1️⃣ **(3 + 5 * 2 + 8 / 3) - 7 / 2**

2️⃣ **10 + 6 * 3 - 4 / 2**

3️⃣ **(12 / 3) + (8 % 5) * 2**

4️⃣ **5 * 2 - 3 * (6 / 2)**

5️⃣ **(9 - 3) * (4 + 2)**

6️⃣ **18 / 4 + 7 % 3**

7️⃣ **2 * 3 + 10 / 5 - 1**

8️⃣ **(15 - 9 / 3) * 2 + 1**

9️⃣ **20 / 3 + 5 % 2**

🔟 **(8 + 2) * 2 / 5 - 3**

Answer:

1️⃣ **(3 + 5 * 2 + 8 / 3) - 7 / 2**

Solution:
5 * 2 = 10
8 / 3 = 2
3 + 10 + 2 = 15
7 / 2 = 3
15 - 3 = **12**

✅ **Answer: 12**

2️⃣ **10 + 6 * 3 - 4 / 2**

Solution:

6 * 3 = 18

4 / 2 = 2

10 + 18 - 2 = **26**

✅ **Answer: 26**

3️⃣ **(12 / 3) + (8 % 5) * 2**

Solution:

12 / 3 = 4

8 % 5 = 3

3 * 2 = 6

4 + 6 = **10**

✅ **Answer: 10**

4️⃣ **5 * 2 - 3 * (6 / 2)**

Solution:

5 * 2 = 10

6 / 2 = 3

3 * 3 = 9

10 - 9 = **1**

✅ **Answer: 1**

5️⃣ **(9 - 3) * (4 + 2)**

Solution:

9 - 3 = 6

4 + 2 = 6

6 * 6 = **36**

✅ **Answer: 36**

6️⃣ **18 / 4 + 7 % 3**

Solution:
18 / 4 = 4
7 % 3 = 1
4 + 1 = **5**

✅ **Answer: 5**

7️⃣ **2 * 3 + 10 / 5 - 1**

Solution:
2 * 3 = 6
10 / 5 = 2
6 + 2 - 1 = **7**

✅ **Answer: 7**

8️⃣ **(15 - 9 / 3) * 2 + 1**

Solution:
9 / 3 = 3
15 - 3 = 12
12 * 2 = 24
24 + 1 = **25**

✅ **Answer: 25**

9️⃣ **20 / 3 + 5 % 2**

Solution:
20 / 3 = 6
5 % 2 = 1
6 + 1 = **7**

✅ **Answer: 7**

🔟 **(8 + 2) * 2 / 5 - 3**

Solution:

8 + 2 = 10

10 * 2 = 20

20 / 5 = 4

4 - 3 = **1**

✅ **Answer: 1**

Question 2: Given the following program, show the corresponding output on the screen *(assume that the user entered age = 16)*

```
#include <iostream>

using namespace std;

int main()

{

   int age;

   cout << "Enter your age: ";

   cin >> age;

   cout << endl;

   if (age >= 18)

      cout << "You are eligible to vote." << endl;

   else

      cout << "You are not eligible to vote yet." << endl;

   return 0;

}
```

Question 3: Given the following program, show the corresponding output on the screen *(assume that the user entered temp = 35)*

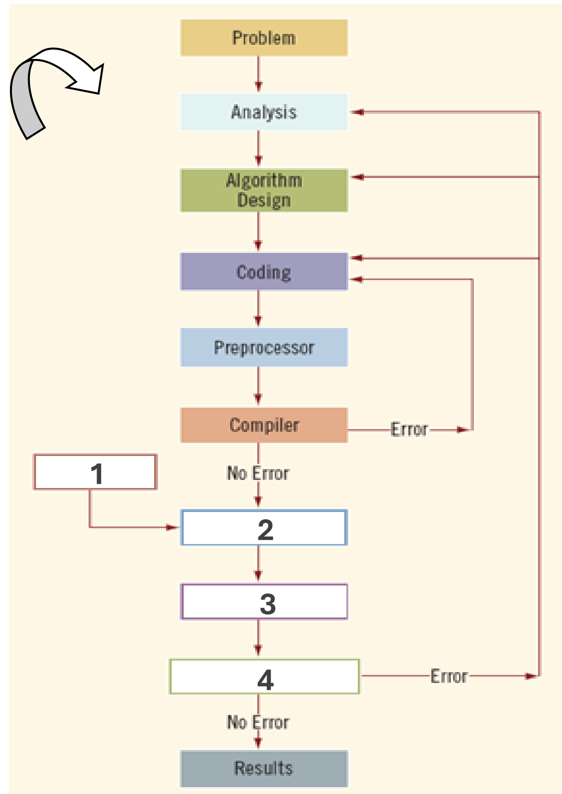
```
#include <iostream>

using namespace std;

int main()

{

   int temp;

   cout << "Enter the temperature: ";

   cin >> temp;

   cout << endl;

   if (temp > 40)

      cout << "It is a very hot day!" << endl;

   else if (temp >= 25)

      cout << "The weather is warm." << endl;

   else

      cout << "It’s a cool day." << endl;

   return 0;

}
```

Question 4: Given the following program, show the corresponding output on the screen (assume that the user entered gpa=1.8)

```
#include <iostream>
using namespace std;

int main()
{
double gpa;

cout << "Enter the GPA: ";
cin >> gpa;
cout << endl;

if (gpa >= 3.9)
        cout << "Dean\'s Honor List." << endl;
else if (gpa < 2.0)
        cout << "The GPA is below the graduation "
        << "requirement. \n See your "
        << "academic advisor." << endl;

return 0;
}
```

Question 5:

i. **Given the following diagram for the Problem Analysis–Coding–Execution Cycle, complete the following lines to show the details of the machine-role in that cycle:**



1. ....
2. ....
3. ....
4. ....

Question 6: Given the following program, show the corresponding output on the screen:

```
#include <iostream>
using namespace std;
 int main()
{
  int x=13, y = 24, z=5;
  double w=3.0;

  z = x--;
  w = x/z+y;
  cout << "Value of Z is: " << z << endl ;
  cout << "Value of W is: " << w << endl ;
  if (z>5)
    cout << "Last Value of Z is: " << --z << endl ;

  return 0;
}
```

Question 7: **C++ Coding**

1️⃣ Simple Sum

Write a C++ program that asks the user to enter two numbers, then displays their **sum**.

2️⃣ Average of Three Numbers

Write a program that reads **three numbers** from the user and calculates their **average**.

Hint: average = (num1 + num2 + num3) / 3

3️⃣ Rectangle Area

Write a program that asks the user to enter the **length** and **width** of a rectangle, then prints the **area**.

Hint: area = length * width

4️⃣ Convert Temperature

Write a program that reads a temperature in **Celsius** and converts it to **Fahrenheit** using the formula:
F = (C * 9 / 5) + 32

5️⃣ Simple Expression

Write a program to evaluate and display the result of this expression:
(3 + 5 * 2 + 8 / 3) - 7 / 2

Hint: use integer division for /.